Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration*	**Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name and Signature of Student and Date

Sheikh.
21/02/26

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Session	2025-26 (EVEN)	Course Name	Competitive Coding Lab
Semester	6	Course Code	23IOT1605
Roll No	63	Name of Student	Sheikh Wasimuddin
Practical Number	DBMS SLOT-1		

Dataset01:Employee Database

EmployeeTable

emp_id	emp_name	dept_id	salary	manager_id	join_date
1	Alice	10	70000	NULL	2020-01-10
2	Bob	20	60000	1	2021-03-15
3	Charlie	10	80000	1	2019-07-20
4	David	30	55000	2	2022-06-01
5	Eve	20	65000	2	2020-11-25

DepartmentTable

dept_id	dept_name
10	HR
20	IT
30	Finance

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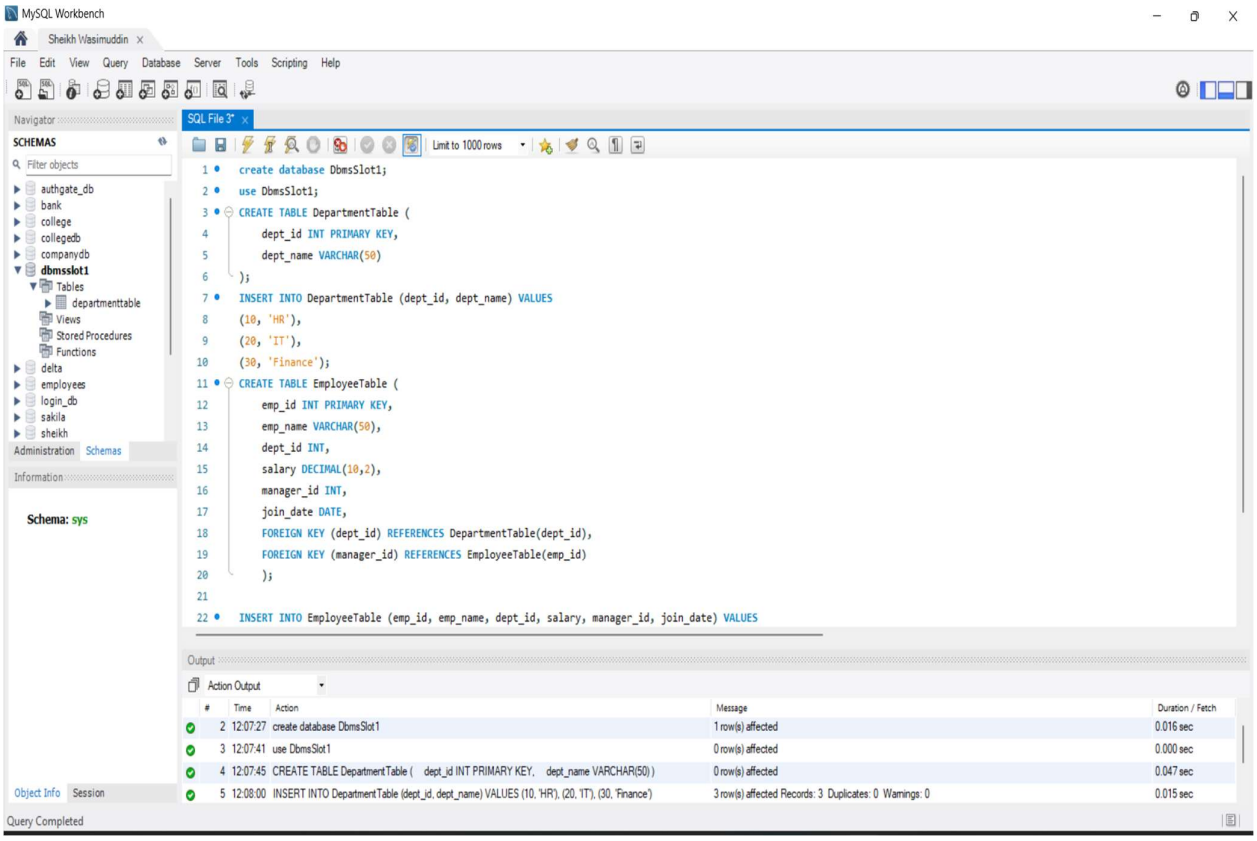

PracticeQuestions:

1. Find employees who earn more than their manager.

```
SELECT e.emp_name, e.salary, m.emp_name AS manager_name, m.salary AS manager_salary
FROM EmployeeTable e
JOIN EmployeeTable m
ON e.manager_id = m.emp_id
WHERE e.salary > m.salary;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	salary	manager_name	manager_salary
Charlie	80000.00	Alice	70000.00
Eve	65000.00	Bob	60000.00

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2. Display the second highest salary.

```
SELECT MAX(salary) AS second_highest_salary
FROM EmployeeTable
WHERE salary < (SELECT MAX(salary) FROM EmployeeTable);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

second_highest_salary
70000.00

3. Find department-wise average salary

```
SELECT d.dept_name, AVG(e.salary) AS avg_salary
FROM EmployeeTable e
JOIN DepartmentTable d
ON e.dept_id = d.dept_id
GROUP BY d.dept_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

dept_name	avg_salary
HR	75000.000000
IT	62500.000000
Finance	55000.000000

4. Fetch employees who joined before their manager

```
SELECT e.emp_name, e.join_date, m.emp_name AS manager_name, m.join_date AS manager_join_date
FROM EmployeeTable e
JOIN EmployeeTable m
ON e.manager_id = m.emp_id
WHERE e.join_date < m.join_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	join_date	manager_name	manager_join_date
Charlie	2019-07-20	Alice	2020-01-10
Eve	2020-11-25	Bob	2021-03-15

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5. Find departments with more than 1 employee

```
SELECT dept_id, COUNT(*) AS total_employees
FROM EmployeeTable
GROUP BY dept_id
HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wra

dept_id	total_employees
10	2
20	2

6. Show the highest-paid employee in each department.

```
SELECT emp_name, dept_id, salary
FROM EmployeeTable e
WHERE salary = (
    SELECT MAX(salary)
    FROM EmployeeTable
    WHERE dept_id = e.dept_id
);
```

Result Grid | Filter Rows: | Expo

emp_name	dept_id	salary
Charlie	10	80000.00
David	30	55000.00
Eve	20	65000.00

7. Find employees who do not have a manager.

```
SELECT emp_name
FROM EmployeeTable
WHERE manager_id IS NULL;
```

Result Grid | Filter Rows:

emp_name
Alice

8. Display employees earning above company average salary.

```
SELECT emp_name, salary
FROM EmployeeTable
WHERE salary > (SELECT AVG(salary) FROM EmployeeTable);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	salary
Alice	70000.00
Charlie	80000.00

9. Write a query to rank employees by salary within each department.

```
SELECT emp_name, dept_id, salary,
       RANK() OVER (PARTITION BY dept_id ORDER BY salary DESC) AS salary_rank
FROM EmployeeTable;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_id	salary	salary_rank
Charlie	10	80000.00	1
Alice	10	70000.00	2
Eve	20	65000.00	1
Bob	20	60000.00	2
David	30	55000.00	1

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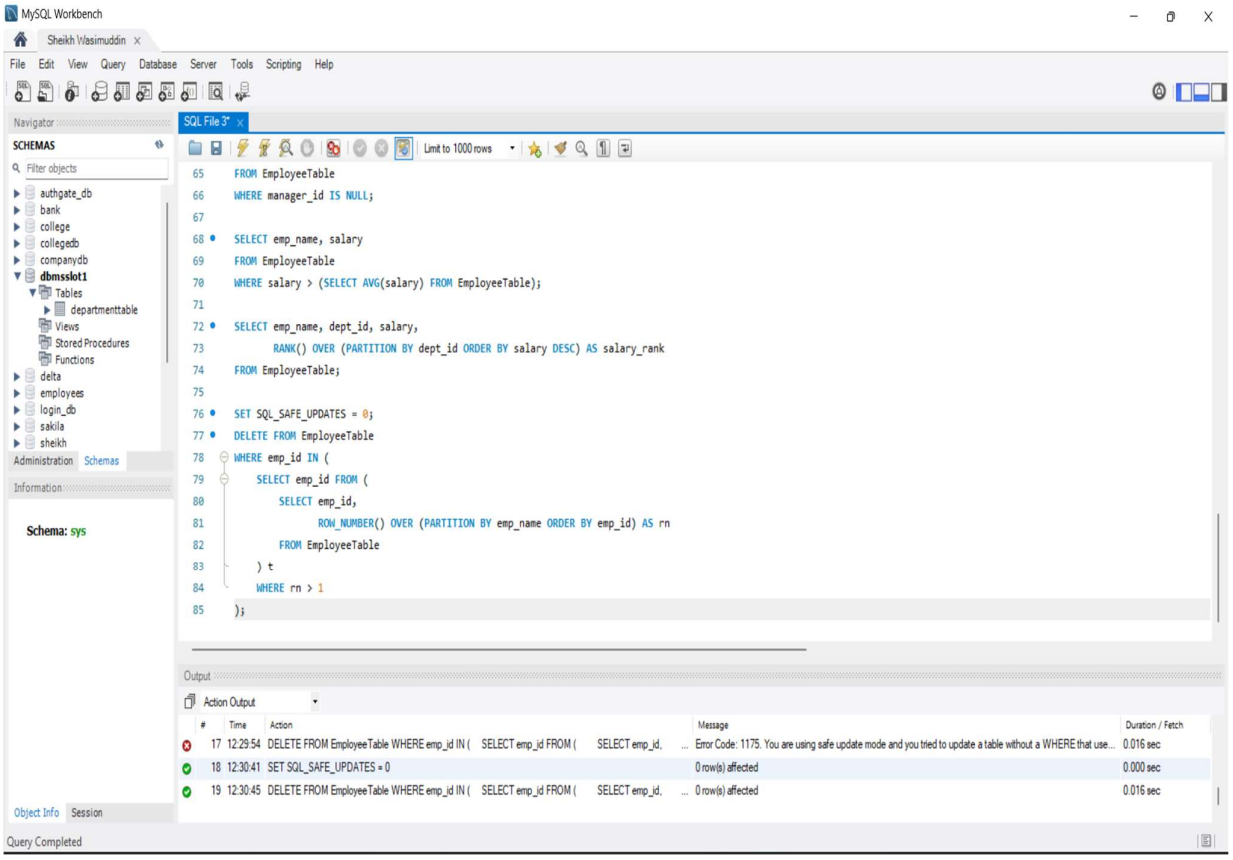
10. Delete duplicate employee records based on emp_name.



Dataset02:StudentMarks

student_id	subject	marks
1	Math	90
1	Science	85
2	Math	70
2	Science	75
3	Math	95
3	Science	88

Practice questions

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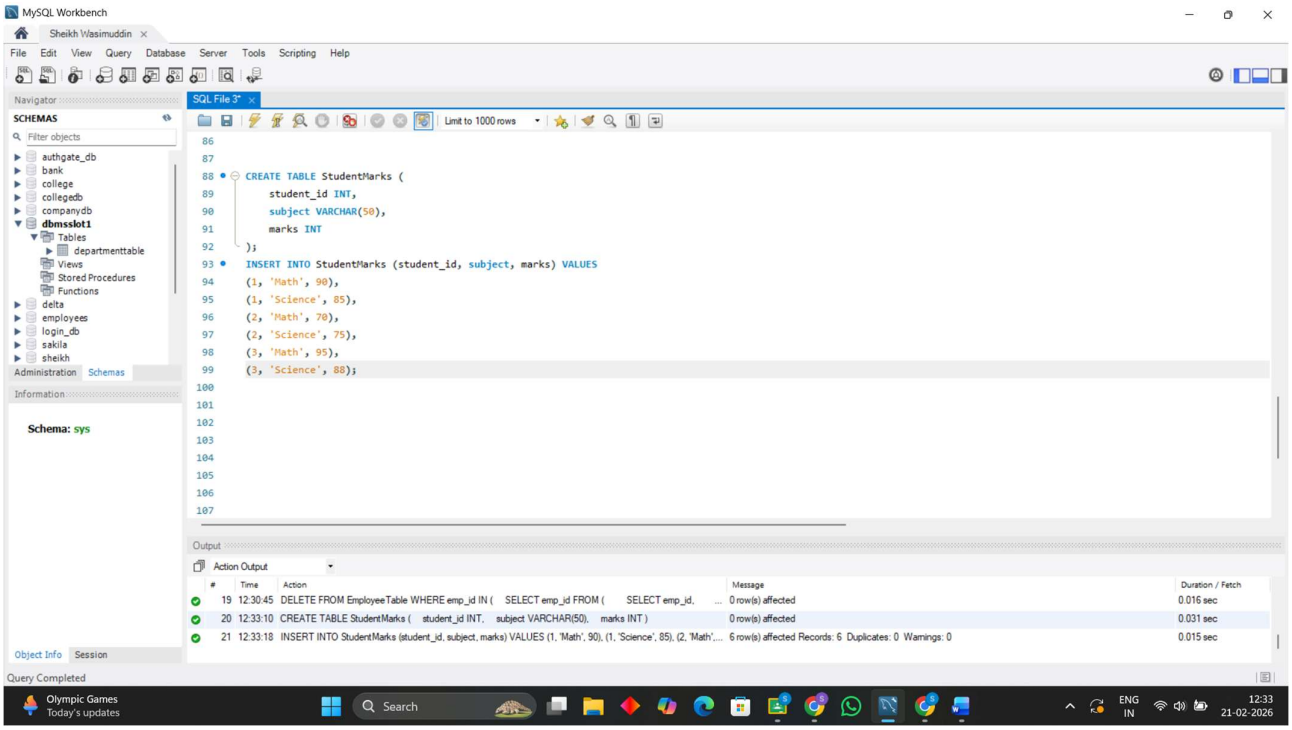

1. Find students who scored above average in every subject.

```
SELECT student_id
FROM StudentMarks s
WHERE marks > (
    SELECT AVG(marks)
    FROM StudentMarks
    WHERE subject = s.subject
)
GROUP BY student_id
HAVING COUNT(*) = (SELECT COUNT(DISTINCT subject) FROM StudentMarks);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id
1
3

2. Find the topper in each subject.

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```
SELECT student_id, subject, marks
FROM StudentMarks s
WHERE marks = (
    SELECT MAX(marks)
    FROM StudentMarks
    WHERE subject = s.subject
);
```

Result Grid | Filter Rows:

student_id	subject	marks
3	Math	95
3	Science	88

3. Calculate overall percentage per student.

```
SELECT student_id,
       SUM(marks) AS total_marks,
       AVG(marks) AS percentage
FROM StudentMarks
GROUP BY student_id;
```

Result Grid | Filter Rows:

student_id	total_marks	percentage
1	175	87.5000
2	145	72.5000
3	183	91.5000

4. Pivot the table to show subjects as columns.

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```
SELECT student_id,
       MAX(CASE WHEN subject = 'Math' THEN marks END) AS Math,
       MAX(CASE WHEN subject = 'Science' THEN marks END) AS Science
FROM StudentMarks
GROUP BY student_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	Math	Science
1	90	85
2	70	75
3	95	88

5. Find students who scored same marks in all subjects.

```
SELECT student_id
FROM StudentMarks
GROUP BY student_id
HAVING MIN(marks) = MAX(marks);
```

Result Grid | Filter Rows:

student_id